

Balaji Bansude

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Summary

Aspiring Data Scientist with expertise in Machine Learning, Deep Learning, NLP, and Generative AI. Passionate about developing scalable AI solutions using Python, FastAPI, LangChain, RAG, and data-driven analytics.

Technical Skills

Programming: Python, SQL, JavaScript

Data Analysis: NumPy, Pandas, Matplotlib, Seaborn, Data Cleaning, EDA

Machine Learning: Scikit-learn, XGBoost, Random Forest, Regression, Classification, Model Evaluation

Generative AI: LangChain, RAG, OpenAI API, FAISS

Backend: FastAPI, REST APIs

Database: PostgreSQL, MongoDB

Tools: Git, GitHub, Docker, Jupyter Notebook, VS Code

Projects

Enterprise AI Knowledge Assistant

- Reduced document search time by over **85%** by implementing a Retrieval-Augmented Generation (RAG) pipeline across **10,000+ enterprise documents**, enabling instant context-aware responses.
- Automated ingestion and indexing of PDF, DOCX, Excel, and scanned documents using OCR, semantic chunking, and FAISS, eliminating manual knowledge retrieval.
- Engineered a scalable FastAPI backend supporting concurrent users with JWT authentication, conversation memory, role-based access control, and source-cited AI responses.
- Improved knowledge discovery by integrating LangChain, OpenAI models, and vector search to generate accurate summaries, document comparisons, and meeting notes.

Predictive Healthcare AI Assistant

- Built predictive models using XGBoost and Neural Networks to estimate disease risk from patient symptoms and laboratory reports, improving diagnostic accuracy.
- Accelerated medical report analysis by enabling semantic search over **20,000+ clinical records** using LangChain and FAISS, reducing manual review time.
- Generated AI-powered health summaries, medication explanations, and follow-up recommendations through Large Language Models, improving patient understanding.
- Designed a secure multi-user healthcare platform with FastAPI APIs, authentication, patient history management, and scalable report processing.

AI Financial Intelligence Platform

- Built machine learning models including XGBoost, LightGBM, and LSTM to predict stock movements, detect fraud, and assess credit risk using large financial datasets.
- Processed and analyzed financial statements containing **100,000+ transactions**, enabling automated investment insights and business risk assessment.
- Implemented Retrieval-Augmented Generation to summarize annual reports, compare financial statements, and explain key business metrics with cited evidence.
- Developed interactive analytics dashboards delivering real-time forecasting, portfolio monitoring, and AI-driven financial recommendations for multiple users.

Education

MIT ADT University

M.Sc. Artificial Intelligence and Machine Learning

2025 – Present

CGPA: 8.15

Vidya Pratishthan's College, Indapur

B.Sc. Computer Science

2022 – 2025

CGPA: 7.67

Certifications

- IBM Machine Learning
- Full Stack Web Development With MERN Stack and GEN AI